# Appendix B: Technical Notes

## Determinism I

All stochastic steps use fixed seeds (seed $= 42$ for NMF, SVD, and graph layouts). The fixture corpus, TF-IDF/SVD embeddings, and topic factorization are byte-stable across runs. Graph centrality scores are rounded to a fixed precision and ranked with a node-id tiebreaker so that floating-point non-associativity in iterative solvers cannot perturb the reported rankings. Record identity uses a content digest (SHA-1, `usedforsecurity=False`) rather than a salted hash, so de-duplication and corpus byte-stability hold across processes.

# Data Model I

Each record is a `Paper` dataclass with: title, abstract, authors (list of `Author`), year, DOI, arXiv ID, Semantic Scholar ID, OpenAlex ID, venue, citation count, references (list of canonical IDs), publication date, PDF URL, open-access flag, and full-text source. The canonical identifier hierarchy governs de-duplication and citation resolution:

$$\text{canonical_id} = \begin{cases} \texttt{doi:} + \text{normalize}(\text{DOI}) & \text{if DOI present} \\ \texttt{arxiv:} + \text{arXiv_id} & \text{if arXiv ID present} \\ \texttt{s2:} + \text{S2_id} & \text{if S2 ID present} \\ \texttt{openalex:} + \text{OpenAlex_id} & \text{if OpenAlex ID present} \\ \texttt{title:} + \text{SHA1}(\text{title})[:16] & \text{otherwise} \end{cases}$$

DOI normalization lower-cases the DOI and strips any `https://doi.org/` or `dx.doi.org/` prefix, so the same paper returned by two engines under different case or format variants merges to a single canonical ID.

## NMF Mathematics I

Non-negative matrix factorization decomposes the TF-IDF document-term matrix $\mathbf{V} \in \mathbb{R}^{m \times n}$ (where $m$ is the number of documents and $n$ is the vocabulary size) into $\mathbf{W} \in \mathbb{R}^{m \times k}$ and $\mathbf{H} \in \mathbb{R}^{k \times n}$, where $k$ is the number of topics (here 5). The factorization minimizes:

$$\min_{\mathbf{W},\mathbf{H} \geq 0} \|\mathbf{V} - \mathbf{W}\mathbf{H}\|_F^2$$

using multiplicative update rules (Lee and Seung 1999) with a fixed random seed for reproducibility. The topic-term matrix $\mathbf{H}$ gives the top terms per topic; the document-topic matrix $\mathbf{W}$ gives each document's topic loadings.

## Growth Rate Estimation I

The compound annual growth rate is:

$$\mathrm{CAGR} = \left( \frac{N_{\mathrm{end}}}{N_{\mathrm{start}}} \right)^{1/(T_{\mathrm{end}} - T_{\mathrm{start}})} - 1$$

where $N_{\mathrm{start}}$ and $N_{\mathrm{end}}$ are the publication counts in the first and last years of the corpus, respectively. The doubling time is $t_d = \ln(2) / \ln(1 + \mathrm{CAGR})$. For this run: CAGR $= 6.76\%$, doubling time $= 2.1$ years.

## Configuration Surface I

A single `manuscript/config.yaml` controls the search term, per-engine query and keyword sets, engine enable toggles, subfield taxonomy, hypotheses, full-text and embedding options, and paper metadata. This run drew on 4 engines, a 4-bucket taxonomy, and 4 hypotheses.

# Artifacts I

Intermediate and final outputs live under `output/` and are disposable and regenerable:

# Artifacts I

| File | Stage | Description |
|---|---|---|
| `corpus.jsonl` | 01 | De-duplicated corpus (46 records) |
| `temporal_analysis.json` | 02 | Year counts, CAGR, doubling time, peak year |
| `subfield_classification.json` | 02 | Per-bucket paper counts |
| `subfield_timeline.json` | 02 | Per-subfield annual breakdown |
| `tfidf_data.json` | 02 | TF-IDF matrix, feature names, doc tokens |
| `topics.json` | 02 | NMF topic-term distributions |
| `citation_network.json` | 02 | Network metrics, PageRank, HITS, communities |
| `citation_graph.gml` | 02 | GraphML citation graph |

# Artifacts I

| File | Stage | Description |
|---|---|---|
| `nanopublications.jsonl` | 03 | LLM-extracted assertions (0 in this run) |
| `hypothesis_scores.json` | 03 | Per-hypothesis evidence scores |
| `cross_phase_analysis.json` | 01 / 03 | Phase membership, overlap, citation sufficiency, and explicit hypothesis-score claim boundary |
| `fulltext_assessment.json` | 06 | Abstract/OA/PDF coverage report |